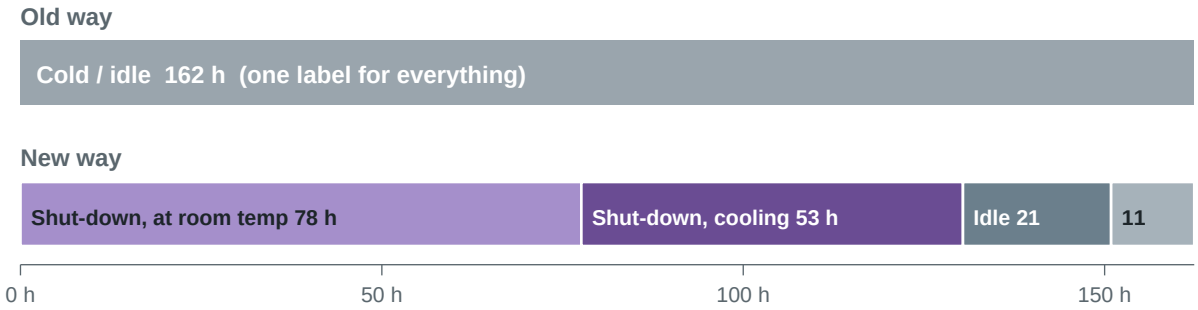
Idle and Shut-down, sorted out

Contents: 1. What changed · 2. How the rule decides · 3. What it did to the numbers

1. What changed

When the plant is switched off, every minute now gets one of two clear names. If the stack is still warm and nothing shut it down, it is **Idle**. If it is close to room temperature, or still cooling after a shut-down, it is **Shut-down**.

Before, all of these hours sat in one bucket called *Cold / idle*. That bucket mixed a stack at 20 °C with a stack at 260 °C. A model learning from the data could not tell a plant that is resting from one that is still cooling down. Splitting the bucket fixes that.



Idle = 21 h after a stopped heat-up + 11 h in the first data file. Drawn to scale.

The 162 hours the plant was off, across all 39 logged days. Same hours, same data; only the labels changed.

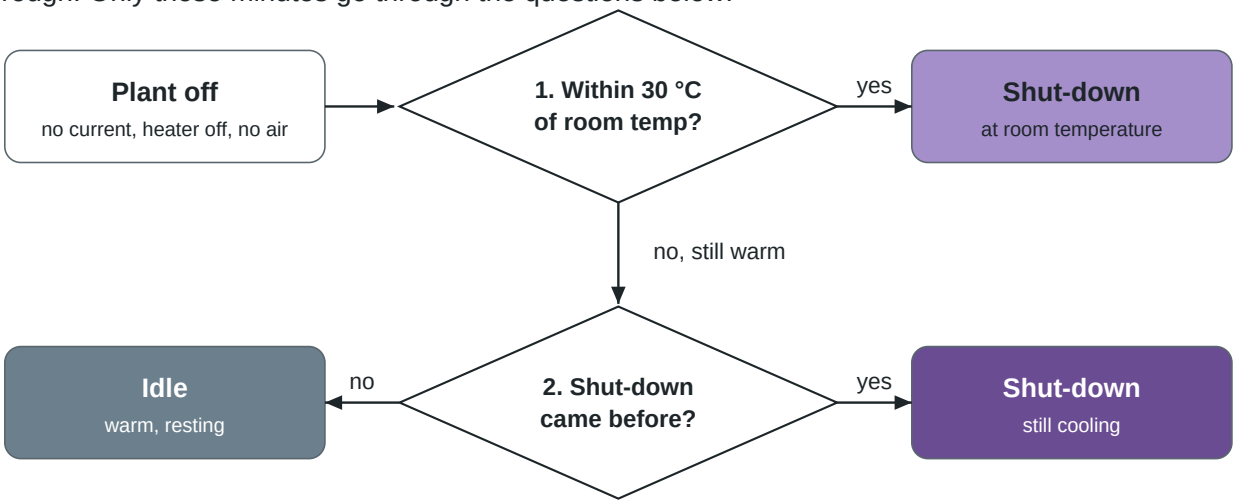
Idle	Plant off, stack still warm, and no shut-down came before it. Example: a heat-up that was stopped halfway.
Shut-down, still cooling	Plant off after a shut-down. The stack loses heat slowly, about 1–2 °C per hour.
Shut-down, at room temp	Plant off and the stack is within 30 °C of the room. Nothing more is happening.

If you remember one thing: “Idle” now means warm and resting with no shut-down before it. Anything near room temperature counts as Shut-down.

2. How the rule decides

The rule asks two questions about every minute the plant is off.

“Off” means three things at once: no electric current from the stack, the air heater off and no air blowing through. Only those minutes go through the questions below.

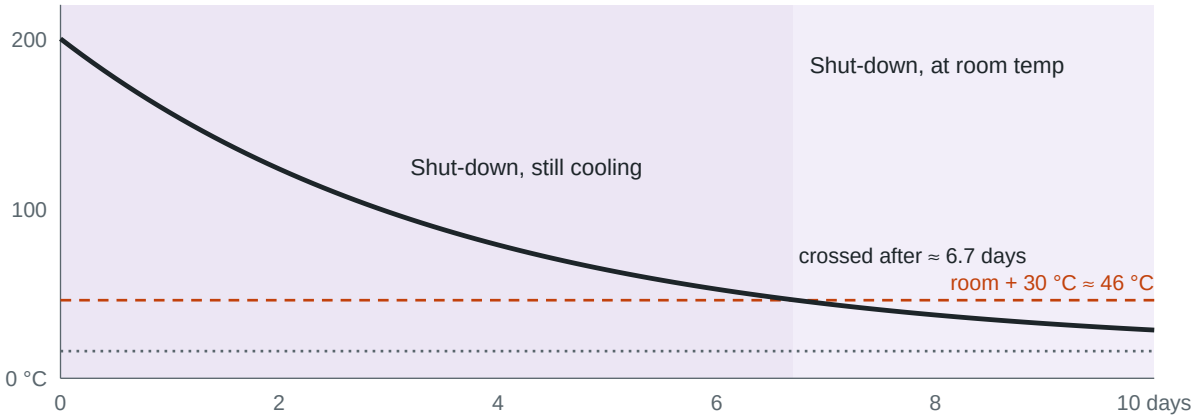


Question 2 also looks back across midnight and up to 7 days of missing files.

Every plant-off minute goes through question 1 first, then question 2.

What “room temperature” means here

The plant has no room-temperature sensor. The rule uses four probes on the cold supply pipes of the fuel module as a stand-in. They read between 10 and 18 °C. So “within 30 °C of room temperature” means the stack is below roughly 40–48 °C.

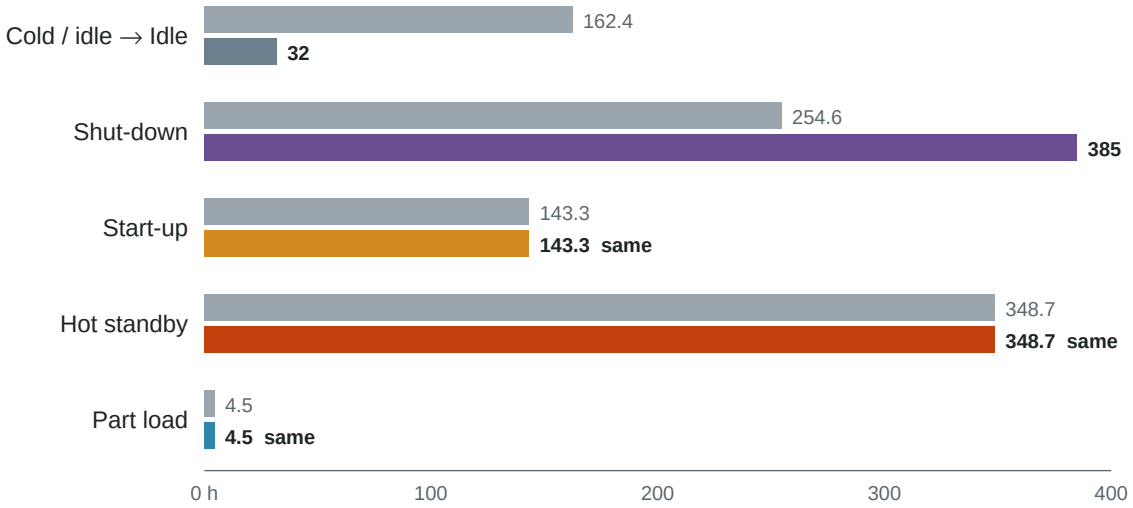


A stack left alone after a shut-down at 200 °C, with heater and air off. It cools like a cup of tea: fast at first, then slower (measured cooling time constant about 89 hours). The label switches when the curve drops below the dashed line.

If you remember one thing: the rule needs only the stack temperature, a stand-in for room temperature, and what happened just before.

3. What it did to the numbers

Only the plant-off hours were renamed. Everything else stayed exactly as before.



Hours per stage. Grey bar = before, coloured bar = after. 39 logged days, about 913 hours. The 130 hours that left the old bucket all went to Shut-down. Rated power never occurs in the data.

Check	Before	After	What it means
Consistency checks that raised a flag	37	24	The 9 clashes with the old “below 50 °C” rule are gone. The rest are older, known findings.
Signal-rebuild scores and test-day groups	—	identical	The learning library was rebuilt; only its labels changed.
Cooling 200 °C → room + 30 °C	—	≈ 6.7 days	A Shut-down that starts warm usually ends with the next start-up.

Still open

• **30 °C or 40 °C margin?** On 26 and 28 March the room stand-in read only about 10 °C, while the stack sat flat at 38–47 °C. With 30 °C those 19 hours count as “still cooling”; with 40 °C they count as “at room temperature”. Both are Shut-down either way.

• **A real room-temperature sensor.** The stand-in probes sit in a heated space. A container or outdoor temperature tag from the plant documents would replace them.

Where it lives: rule in data_analysis/operating_states.py · definition in operational-modes.md §1 · full study in data_analysis/cold_idle_analysis.ipynb

If you remember one thing: no hours were added or lost. 130 hours moved from the old “Cold / idle” bucket to Shut-down, and 32 hours became Idle.